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Effects of fungicides on decomposer communities and litter decomposition in vineyard streams

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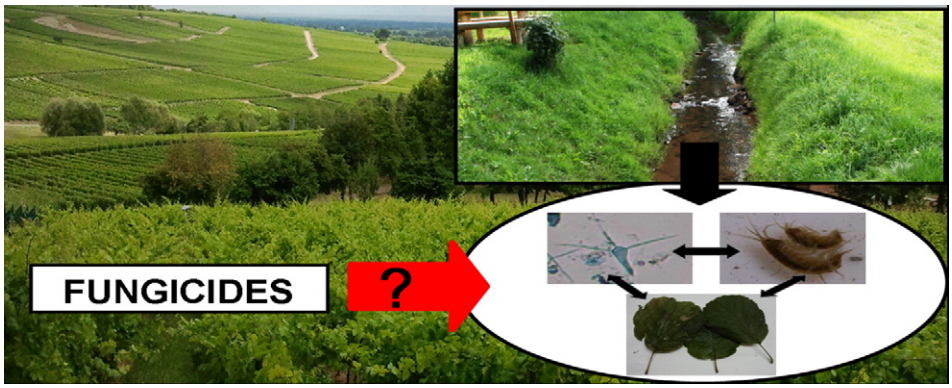
HIGHLIGHTS

- We conducted a field study in 17 fungicide-exposed vineyard streams.
- Microbial and shredder communities changed along the fungicide toxicity gradient.
- Microorganism-mediated decomposition decreased up to 40% in polluted streams.
- Invertebrate-mediated decomposition was negatively associated with sediment copper.
- Additional laboratory experiment suggested causality of fungal community changes.

GRAPHICAL ABSTRACT



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ABSTRACT

Large amounts of fungicides are applied globally and partly enter freshwater ecosystems. A few laboratory studies examined their effects on decomposer communities and the ecosystem process of litter decomposition (LD), whereas the field situation remains largely unknown. We conducted a field study with 17 stream sites in a German vineyard area where fungicides represent the dominant pest control agent. Passive samplers were used to monitor 15 fungicides and 4 insecticides in streams and their toxicity was described using the toxic unit approach, whereas sediment samples were taken to characterise total copper concentrations. Microbial and leaf-shredding invertebrate community composition and related LD rates were assessed at each site. The structure of microbial and shredder communities as well as fungal biomass changed along the fungicide toxicity gradient. The changes in microbial endpoints were associated with a reduction of microbial LD rate of up to 40% in polluted streams. By contrast, neither the invertebrate LD rate nor in-situ measured gammarid feeding rates correlated with fungicide toxicity, but both were negatively associated with sediment copper concentrations. A subsequent laboratory experiment employing field fungicide concentrations suggested that the microbial community changes are causal. Overall, our results suggest that fungicides can affect LD under field conditions.
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1. Introduction

Human activities exert high pressure on freshwater ecosystems. Among the different stressors, pollution has been identified as a major driver of freshwater biodiversity loss (MEA, 2005; Vörösmarty et al.,

Abbreviations: LD, litter decomposition; TU, Toxic Units; SumTU, logarithmic sum of Toxic Units; *k*, leaf decomposition rate.

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2010) consequently threatening ecosystem health and the provisioning of ecosystem processes and services (Dudgeon et al., 2006). In this context, pesticides contribute substantially to freshwater pollution because large amounts are applied worldwide and partly enter surface waters (Köhler and Triebskorn, 2013; Schwarzenbach et al., 2010) with rainfall-triggered runoff as a major input path (Leu et al., 2004a). When entering a stream, they may affect the different groups of organisms involved in allochthonous coarse litter decomposition (LD) which in turn might propagate to a reduction of this ecosystem process (Peters et al., 2013; Schäfer et al., 2007). LD plays a key role in stream ecosystems, because it represents the dominant energy and nutrient source for the heterotrophic food web (Wallace et al., 1997), particularly in headwaters (Fisher and Likens, 1973). In addition, dominance of allochthonous inputs over primary production can extend up to tens of kilometres from headwaters (Webster, 2007). Considering that 50% of the approximately 1.2 million km of the stream network (scale 1:250,000) in Europe are small rivers and streams (Globevnik, 2007) and that heterotrophic metabolism dominates in low-order streams (Vannote et al., 1980), allochthonous inputs are of central importance. In this context, fungi and bacteria are the main microbial decomposers converting organic matter into a more nutritious food resource for leaf-shredding macroinvertebrates (Gessner et al., 2007). Despite the fact that fungicides are the most heavily used group of pest control agents in regions such as the European Union and that they can affect non-target freshwater fungi (Dijksterhuis et al., 2011), they represent the least studied group of pesticides (Köhler and Triebskorn, 2013). Under laboratory conditions, fungicides change fungal community structure and decrease fungal biomass (Bundschuh et al., 2011b), which resulted in a decrease in microbial LD (Artigas et al., 2012; Rasmussen et al., 2012a). Moreover, shredders showed preference for non-exposed leaves in comparison to leaves that have been exposed to fungicides (Bundschuh et al., 2011b) and were directly affected by fungicides at environmentally relevant concentrations (Flores et al., 2014; Zubrod et al., 2014). However, it remains open whether and to which extent these studies can be extrapolated to the field situation, where multiple factors such as biotic interactions, abiotic factors and recolonisation may moderate the effects of fungicides on LD.

In this study we assessed the effects of fungicides on LD and the associated decomposer communities in 17 streams within a vineyard area where fungicides are the dominant pest control agent. Microbial and leaf-shredding invertebrate communities were characterised together with microbial and invertebrate-mediated LD in autumn 2012 (Supplementary data Fig. S1), coinciding with the major leaf litter input into streams. Fungicide exposure in stream water was monitored the same year during four rainfall events in summer and early autumn, covering the main fungicide application period. At the same time, the ecotoxicological potential of the pesticide loads introduced during these run-off events was empirically estimated by means of in-situ bioassays using gammarid feeding as endpoint. Moreover, the toxicity of each pesticide for microorganisms and invertebrates at the different sampling sites was assessed using Toxic Units (Ohe and Zwart, 2013) and subsequently aggregated per site and rainfall event. Total copper in stream sediment was also assessed because of its use as fungicide in vineyards, where it tends to accumulate in the soil and reaches the stream mainly via physical erosion (Bereswill et al., 2012). Finally, a laboratory experiment was conducted to identify whether correlations between microbial community change and fungicide toxicity are causal. We hypothesised that fungicide toxicity would lead to a shift in the fungal community and a decrease in fungal biomass and that these changes would be associated with a reduction in microbial and invertebrate LD.

Table 1
Environmental variables characterising the 17 sampling sites included in this study.

Variable	Minimum	Maximum	Median	Mean	SD
Stream width (m)	0.80	7.30	1.67	2.21	1.61
Stream depth (m)	0.07	0.43	0.15	0.19	0.10
Current velocity (m/s)	0.01	0.67	0.23	0.26	0.17
Temperature (°C)	11.21	13.77	12.62	12.50	0.81
pH	7.51	8.26	7.87	7.85	0.24
Oxygen (mg/L)	5.30	10.61	9.60	9.10	1.30
Conductivity (μS/cm)	110	1290	332	481	340
Nitrite (mg/L)	0.00	0.80	0.04	0.09	0.19
Nitrate (mg/L)	2	60	5	9	14
Phosphate (mg/L)	0.10	0.60	0.20	0.25	0.13
Ammonium (mg/L)	0.00	0.20	0.00	0.01	0.05
Riffle sections (%)	0	100	80	70	36
Pool sections (%)	0	100	20	30	36
Leaves and wood (<10 cm diameter)[a]	1	3	1	–	–
Wood (>10 cm diameter)[a]	1	2	1	–	–
Filamentous algae[b]	0	2	0	–	–
Macrophytes[b]	0	2	0	–	–
Shading[b]	2	5	4	–	–
WRF (width of the riparian forest; m)[c,d]	1	7.5	2.25	–	–
Tree cover along the bank[b,d]	0.5	5	2.5	–	–
Large substrates (boulder + cobble; %)	0	100	20	31	31
Medium substrates (pebble + gravel; %)	0	60	5	19	23
Fine substrates (sand + silt; %)	0	100	35	50	39
$SumTU_{P.\ subcapitata}$	−4.87	−2.02	−3.73	−3.66	1.07
$SumTU_{D.\ magna}$	−5.34	−1.87	−3.86	−3.91	1.25

[a] Measured using an ordinal scale indicating the coverage, ranging from 1 (<5%) to 3 (>20%).
[b] Measured using an ordinal scale indicating the coverage, ranging from 0 (absent) to 5 (very high).
[c] Width of the riparian forest. When riparian forest was connected with the hillside forest, a maximum of 10 m was assigned.
[d] Mean from the left and right banks.

2. Methods

2.1. Study area

The study was conducted in the wine-growing area of Palatinate (southwest Germany, Supplementary data Fig. S2), which covers more than 23,000 ha of vineyards (Statistisches Landesamt RLP, 2011). Fungicides are applied every 10–14 days from end of April to mid August and are the most used pesticides for grapes (96% of all applications), whereas herbicides (1.5%) and insecticides/acaricides (2.5%) play a minor role (Roßberg, 2010), the latter ones owing to the use of pheromone traps. All streams originate in the Palatinate Forest Nature Park, a forested low-mountain range. After the forest, they discharge through an agricultural landscape that is dominated by vineyards in the first kilometres. We selected 17 sampling sites in different streams covering a presumed gradient of fungicide exposure based on the proportion of vineyards in the upstream catchment (Supplementary data Fig. S2). Four sites were located upstream from vineyards in the forest and served as reference sites without exposure. Sites were selected to exclude other major pressures upstream (such as large waste water treatment plants, large urban areas and industrial facilities) to allow for an identification of potential impacts from fungicides. Moreover, the sampling sites were located within a maximum distance of four km to the forest edge to allow for potential recovery of the invertebrate community from previous effects (Hatakeyama and Yokoyama, 1997; Liess and Ohe, 2005).

2.2. Environmental parameters

Physico-chemical data was collected in concert with leaf deployment at each site using the Rapid Bioassessment Methodology for Rivers and Streams (EPA, 2003; for parameters see Table 1). Nutrient concentrations were analysed on-site with Visocolor® test kits (Macherey-Nagel, Düren, Germany). Water temperature, pH, electrical conductivity and dissolved oxygen were measured using a multiparameter analyser

Multi 340i (WTW, Weilheim, Germany) and flow velocity was measured with a flow meter (Höntzsch, Waiblingen, Germany).

2.3. Fungicide exposure and toxicity

Episodic fungicide exposure was assessed using passive sampling during four precipitation events (>10 mm/day) in summer 2012. Passive samplers were equipped with naked Empore™ SDB-RPS discs and deployed preceding forecasted rainfall events, given that runoff represents a major input path in small agricultural streams (Leu et al., 2004b). They were retrieved within 2 days after the end of the precipitation event, extracted and analysed for 15 fungicides and 4 insecticides (Supplementary data Table S1). The compounds were selected based on information from a previous study in this region (Bereswill et al., 2012) and spraying recommendations from local authorities (www.dlr.rlp.de). Subsequently, time-weighted average concentrations were determined. For further details see Fernández et al. (2014) The toxicity of the pesticide exposure for microorganisms and invertebrates was assessed using Toxic Units ($TU = C_i/EC50_i$, where C_i is the concentration of the pesticide i and $EC50_i$ is the median acute effect concentration for a standard test species for this substance). The algae *Pseudokirchneriella subcapitata* (72-h EC50) and the invertebrate *Daphnia magna* (48-h EC50) were used as proxies for the mixture toxicity for aquatic fungi and macroinvertebrates, respectively. The selection of an algal species as a surrogate for fungi is motivated by previous studies showing that in the absence of toxicity data for fungal species TU based on *P. subcapitata* allowed to establish a relationship with functional microbial community composition as well as LD (Peters et al., 2013; Schäfer et al., 2012a). Toxicity data were compiled from the Pesticide Properties Database (Agriculture and Environment Research Unit of the University of Hertfordshire, 2013). To aggregate the toxicity from different pesticides the logarithmic sum of TU (sumTU) was used (see Supplementary data Table S2 for rationale):

$$sumTU = \log\left(\sum_{i=1}^{n} \frac{c_i}{EC50_i}\right). \quad (1)$$

Sites where no pesticide was detected were assigned a sumTU corresponding to 1/10 of the minimum TU observed in the sites with measured concentrations (−4.9 for $sumTU_{P.\ subcapitata}$ and −5.3 for $sumTU_{D.\ Magna}$). The pesticide toxicity from different rainfall events in each site was aggregated using the maximum sumTU (for rationale see Schäfer et al., 2011). The aggregated toxicity exhibited stronger associations with the biotic endpoints than that related to individual events, suggesting that it is most indicative of the pollution state of a site.

To assess total copper, three sediment samples were taken from the stream bed at each site using a sediment sampler (260 mL volume; Kersting GmbH, Brilon, Germany). Subsequently the three samples from one site were mixed, resulting in a single composite sample. Only one composite sample was taken at each site during the passive sampling period, as copper sediment concentrations can be assumed to be relatively stable compared to water concentrations of polar organic pesticides, due to its accumulation over time periods that substantially exceed discrete rainfall events. In the laboratory, samples were dried at 105 °C, sieved (2-mm mesh size), ground using agate grinding balls and extracted using microwave-induced (MARS Xpress, CEM GmbH) reversed aqua regia (3 HNO3 + HCl) digestion. The extracts were measured with inductively coupled plasma mass spectrometry (ICP-MS, Thermo Fisher Scientific Q-ICP-MS XSeries2).

2.4. Leaf decomposition

Five replicates of coarse-mesh and fine-mesh leaf bags (8 mm and 250 μm mesh size, respectively) were filled with 3 (±0.07 SD; n = 180) g of dried (at 60 °C for 24 h) alder leaves (*Alnus glutinosa*) collected in October 2011 at the time of abscission from a location close to Landau in der Pfalz, Germany (49° 12′ 7″ N; 8° 8′ 37″ E) and stored at −21 °C. Leaf bags were submerged at each sampling site for 21 days starting in mid September to assess for the microorganism and invertebrate-mediated LD, respectively. In addition, temperature loggers were deployed (HOBO Pendant, Synotech, Hückelhoven) to allow for temperature-standardisation of decomposition rates among sites. After retrieval, the remaining leaf material was gently rinsed to remove mineral particles, oven-dried at 60 °C for 24 h and weighed to the nearest 0.001 g. At one of the sampling sites additional replicates of each bag type (n = 5) were returned to the laboratory immediately after immersion and also after 24 h of immersion to determine handling and leaching losses, respectively. The leaf decomposition rate k per sum of degree days ($ddays^{-1}$) was estimated for each site i using the following formula (Benfield, 2007):

$$k = \frac{-\ln\left(\frac{S_i(t)}{S_i(0)}\right)}{\sum_{j=1}^{t} \overline{T}_i(j)} \quad (2)$$

where S is the leaf mass as a function of deployment time t and $\overline{T}$ is the mean temperature for a day j. $S_i(t)$ was corrected for handling and leaching losses. S was assessed from fine-mesh bags when calculating $k_{microbial}$, whereas for $k_{invertebrates}$ $S_i(0)$ was the initial mass in the coarse-mesh bags and $S_i(t)$ was the sum of remaining leaf mass in the coarse-mesh bag and the mean mass loss in the fine-mesh bags at site i.

2.5. Microbial and macroinvertebrate community characterisation

Three additional coarse-mesh leaf bags were deployed at each site to sample the decomposer communities (shredders and microorganisms). These bags were retrieved one week before those used for the assessment of leaf decomposition, namely two weeks after deployment, when more leaf material was remaining to analyse microbial endpoints. The latter include: fungal community structure, fungal biomass and bacterial density. In case of insufficient leaf material, fungal biomass and diversity were prioritised over bacterial density. The bags were gently retrieved using a net (250 μm) to prevent the loss of invertebrates feeding on the leaves. The invertebrates were preserved in 70% ethanol and identified to genus or species level (Bährmann, 2011; Brohmer et al., 2000).

Fungal community structure was determined via conidia identification as described by Bärlocher (1982). Three leaf discs were cut randomly from the remaining leaf material and placed individually in a six-well-plate containing 4 mL of distilled water per well. After incubation for 96 h in darkness on an orbital shaker at 55 rounds per minute, 0.5 mL of lactophenol cotton blue were added to fix samples and to stain the spores. Subsequently, conidia were identified (Ingold, 1975) and counted under a light microscope at 100-fold magnification. For highly abundant spores, only the first 500 were counted. For each sampling site, the mean abundance from the three replicates for each species was used in analysis. Leaf-associated fungal biomass was estimated from ergosterol content, which is a cell–membrane component of eumycotic fungi, following the method of Gessner and Schmitt (1996). Briefly, ergosterol was extracted in alkaline methanol, purified by solid-phase extraction (SPE) and high-performance liquid chromatography and quantified by measuring absorbance at 282 nm. Bacterial density was determined by epifluorescence microscopy as described by Buesing (2005) Briefly, formalin-preserved cells were detached from leaf discs by ultrasonication, filtered on Anodisc membrane filters (Whatman, UK), stained with SYBRGreen II, and automatically counted on digital photographs using an image-analysis system (AxioVision 4.8, Carl Zeiss).

2.6. Feeding rate of gammarids

To assess the ecotoxicological potential of the pesticide loads introduced during run-off events in-situ bioassays using gammarid (*Gammarus fossarum*) feeding were used as measure. Test individuals were collected in a natural stream (Hainbach) near Landau, Germany (49°14′ N; 8°03′ E) and adults of both sexes with a cephalothorax length between 1.2 and 1.6 mm were selected using a passive underwater separation technique (Franke, 1977). These in situ bioassays were involved during two of the four precipitation events for which fungicide toxicity was assessed. Twenty cages (length = 5.0 cm, diameter = 3.0 cm, covered by a 1.0 mm mesh screen) containing one specimen of *G. fossarum* together with two preconditioned (Bundschuh et al., 2011a) leaf discs of known dry weight were placed at each site preceding the rainfall events. Five additional cages that contained only leaf discs were deployed at each site to account for microbial decomposition and abiotic losses. These bioassays can display direct implications of water quality on feeding behaviour. Within 2 days after the end of the rain event, the amphipods together with the remaining leaf material were recovered, dried and weighed. The feeding rate was expressed as consumed leaf mass per mg animal and day and was calculated as described in Maltby et al. (2000).

2.7. Laboratory experiment

A laboratory experiment was performed after the field study to disentangle the influence of two variables (fungicide toxicity and pH) that showed a high association with microbial endpoints in the field. The experiment followed a two-factorial design with minimum and maximum field levels: fungicide exposure (two levels: no fungicides and a mixture of 8 frequently found fungicides at a concentration corresponding to a $\log TU_{P.\ subcapitata}$ of −2 each, resulting in a sumTU of −1.1) and pH (7.5 and 8.3). The mixture included the following fungicides: azoxystrobin, boscalid, cyprodinil, fludioxonil, kresoxim-methyl, metrafenone, pyrimethanil and tebuconazole. A toxicity level of −2 TU was selected because it was frequently found for several fungicides such as azoxystrobin and cyprodinil in the field. We selected a toxicity level in the range of the maximum field toxicity to improve statistical power to detect potential adverse effects, which is usually rather low for experiments such as ours with low replication (see Szöcs and Schäfer, 2015). This toxicity level was used for all fungicides to include potential mixture effects and avoid that responses would only be driven by one compound. Briefly, triplicate 1-L beakers were filled with 800 mL of stream water from a pristine site. Approximately 5 g of alder leaves (wet weight), which were deployed for 10 days at the same pristine site to allow for microbial colonisation, were put in each beaker. The beakers were placed on magnetic stirrers (150 rpm), aerated and kept at room temperature (16 °C) and darkness for 6 days. The temperature was slightly higher than that of field conditions, which should, however, be irrelevant for the main results of our study because it would have affected all treatments in a similar way. The pH was measured daily and adjusted if necessary using sodium hydroxide (2 N) and hydrochloric acid (2 N). After the experiment, the leaf-associated fungal community structure was determined as described above.

2.8. Data analysis

The relationship between fungicide toxicity and both LD and univariate biotic endpoints (fungal biomass and richness, bacterial density and gammarid feeding) was tested using Pearson correlation. Subsequently, linear models were used to identify the environmental variables (including fungicide toxicity) exhibiting the strongest explanatory power for the decomposition rates and the biotic endpoints. Temperature was omitted in the analysis of LD, because LD was temperature-standardised. Fungicide toxicity was included as $sumTU_{P.\ subcapitata}$ for $k_{microbial}$ and as $sumTU_{D.\ magna}$ for $k_{invertebrates}$. To avoid collinearity, predictors were checked for pairwise correlations >0.7 before performing the linear models and the ecologically less relevant variable was removed, which was determined based on expert judgement. Automatic forward model building was performed with the Bayesian Information Criterion (BIC) as stepwise model selection criterion (Schwarz, 1978). The models were checked for error assumptions (normality and homoscedasticity) and unusual observations (leverage, outliers). Finally, the relative importance of retained predictors was assessed using hierarchical partitioning (function lmg in the R package "relaimpo"; Grömping, 2006).

Redundancy analysis (RDA) was used to determine the effects of environmental variables and fungicide toxicity on the community structure of fungi (field study and laboratory experiment) and invertebrates (field study). Collinear variables (Pearson's $r > 0.7$), variables showing minor variation across the study area and rare species occurring in less than 20% of sites and samples from the field and laboratory, respectively, were removed before analysis. Species data were Hellinger transformed to achieve standardisation and circumvent the problems associated with using the Euclidean distance for ecological data (Legendre and Gallagher, 2001). Then automatic model building with forward stepwise model selection was performed (function ordiR2step in the R package "vegan"; Oksanen et al., 2010). Given inflation of the p value in stepwise model building based on hypothesis testing, the p value was set to 0.01 for variable entry. The individual effects from pH and fungicide toxicity in the laboratory experiment were determined using RDA after partialling out the other factor (Peres-Neto et al., 2006). All statistical analyses were conducted in R version 3.1.2 (R Core Team, 2014). Computer code and raw data are provided for reproducibility of the analysis (Supplementary data computer code).

3. Results

3.1. Effects of fungicide toxicity on LD and biotic endpoints

On average, fungicides accounted for 99.2% of the pesticide toxicity in terms of $sumTU_{P.\ subcapitata}$ in polluted sites, except for one site where the insecticide dimethoate occurred in ecotoxicologically relevant concentrations. Still, fungicides accounted for the majority (62.2%) of toxicity in this site. $SumTU_{P.\ subcapitata}$ exhibited statistically significant negative correlations with $k_{microbial}$, fungal biomass (expressed as ergosterol) and fungal richness, whereas the correlation with bacterial density was positive, albeit not statistically significant (Fig. 1). Fungicide toxicity in terms of $sumTU_{D.\ magna}$ did neither correlate with $k_{invertebrates}$ nor the gammarid feeding rate (Supplementary data Fig. S3), but both endpoints correlated close to statistical significance with total copper concentrations in the sediment (Fig. 1). $SumTU_{P.\ subcapitata}$ explained the largest proportion of variance of ergosterol content and fungal richness, whereas physicochemical and habitat variables were better predictors of $k_{microbial}$ and $k_{invertebrates}$, respectively (Table 2).

Copper showed high collinearity with several environmental variables such as conductivity ($r = 0.76$; $p < 0.01$), nitrite ($r = 0.73$; $p < 0.01$) and dissolved oxygen ($r = -0.77$; $p < 0.01$), and moderate collinearity with $sumTU_{D.\ magna}$ ($r = 0.47$, $p = 0.05$). As these chemical variables were relevant for LD in other studies (Pascoal and Cássio, 2004; Schäfer et al., 2012a) they were retained in the linear models whereas copper was omitted. Additional models were performed retaining copper and excluding collinear chemical variables. Water chemistry still presented the highest explanatory power for $k_{microbial}$, whereas for $k_{invertebrates}$ copper explained 32% of the relative explained variance (Supplementary data Fig. S4).

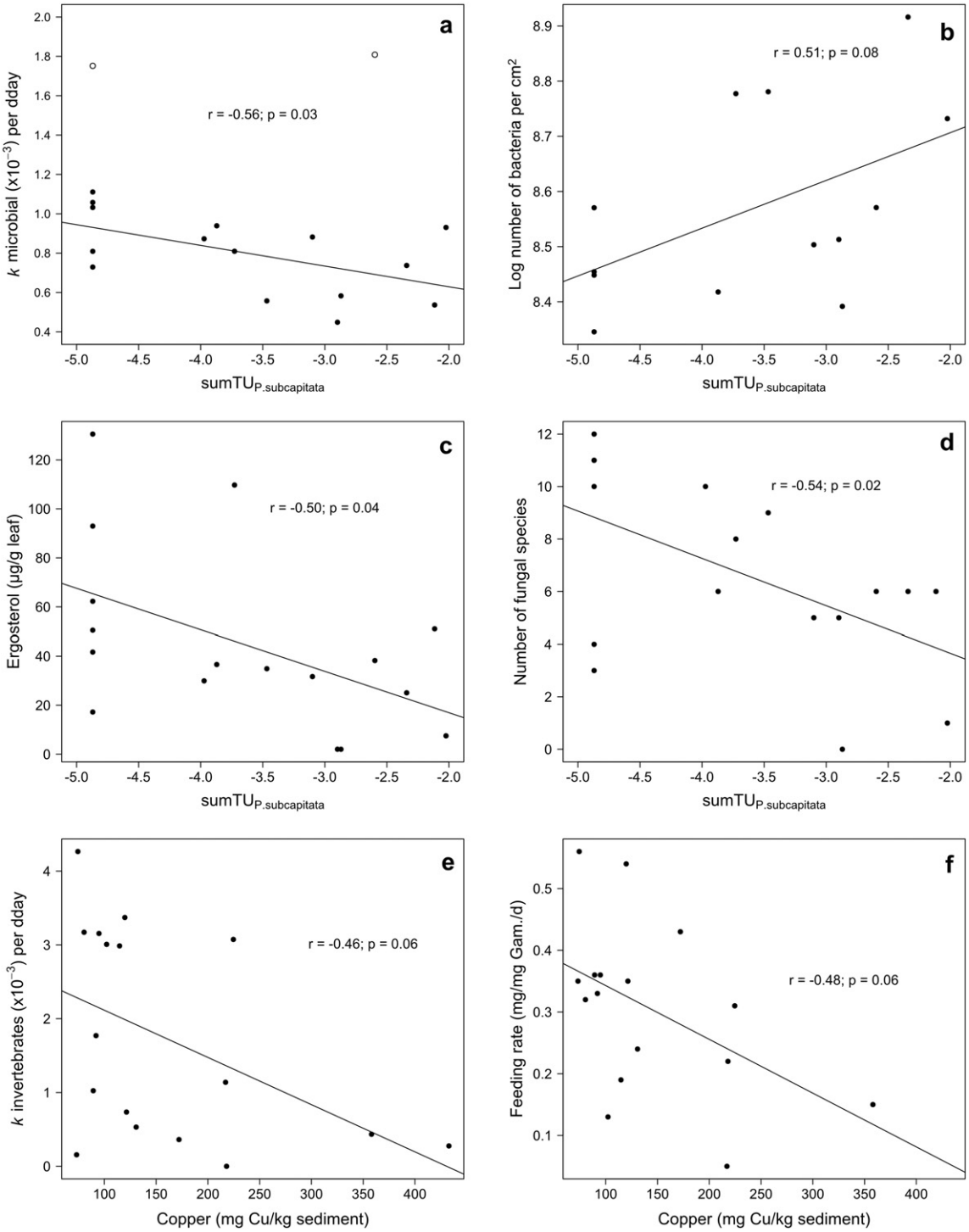

Fig. 1. Correlation between fungicide toxicity and several biotic endpoints. Regression lines are added to visualize the positive or negative correlation. a) Correlation between $k_{microbial}$ and fungicide toxicity in terms of the sum of toxic units for *P. subcapitata* (sumTU) in 17 German streams using time-weighted average concentrations. Empty points in the upper graph (sites 1 and 16) were excluded from the analysis because of undue influence according to Cook's distance. b) Correlation between bacterial density and $sumTU_{P.\ subcapitata}$. Four sites not shown because of insufficient leaf material to determine bacterial density. c) Correlation between fungal biomass expressed as ergosterol and $sumTU_{P.\ subcapitata}$. d) Relationship between the number of fungal species and $sumTU_{P.\ subcapitata}$. e) Correlation between invertebrate leaf decomposition rate and total inorganic copper in stream sediment. f) Relationship between the feeding rate of *G. fossarum* during the third monitored rainfall event and total inorganic copper in stream sediment.

3.2. Effects of environmental variables on the community structure of decomposers

For fungi, only $sumTU_{P.\ subcapitata}$ was a statistically significant predictor for community structure in RDA (Fig. 2a) and explained 12% and 100% of the total and constrained variance, respectively. Conidia abundance of several species increased along the toxicity gradient (e.g. *Anguillospora* spp. and *Lunuluspora curvula*), whereas others showed an inverse trend (e.g. *Tetrachaetum elegans* and *Clavatospora longibrachiata*). Regarding the shredder community, gammarids represented the only shredders in 9 sites and in a further 4 sites they constituted 95% of the shredder community (Supplementary data Fig. S5).

Table 2
Environmental variables (see Table 1 for full names) with highest explanatory power for LD and microbial endpoints. Percentage contribution in hierarchical partitioning, r^2, Bayesian information Criteria (BIC) for the best-fit model and sample size *n* are provided.

Response variable	Explained variance									r^2	BIC	n
	Tree cover	Stream depth	Nitrite	PO4	$SumTU_{P.\ subcapitata}$	pH	Fine substrat.	Medium substrat.	Other env. variables			
$k_{microbial}$[a]			47 (25)			53 (75)				0.71 (0.63)	−267 (−277)	15 (17)
$k_{invertebrates}$	52						48			0.42	−224	17
Ergosterol					100					0.25	121.88	17
Fungal richness		35			65					0.44	41	17
Bacterial density				48	18	9	8	7	9	0.86	−91.28	13

[a] Values in brackets give the result for inclusion of observations that exhibited undue influence according to Cook's distance.

Again, only pesticide toxicity in terms of $sumTU_{D.\ magna}$ was included as explanatory variable of leaf-shredding macroinvertebrate community composition. *Gammarus roeseli* was more abundant in fungicide-polluted streams, whereas *G. fossarum* presented an opposite trend and *Gammarus pulex* showed no response to the fungicide toxicity gradient (Fig. 2b).

For the laboratory experiment, fungicides caused a statistically significant effect on the community structure of leaf-associated fungi ($p = 0.03$), whereas this was not the case for pH after partialling out the fungicide induced effect ($p = 0.27$). The species specific response to toxicity partly deviated from the pattern found in the field (Fig. 2a, c).

4. Discussion

4.1. Effects of fungicides on microbial community and microbial LD

Differences in conidia abundance were found for the sampling sites, which suggests differences in the fungal community composition (Bärlocher, 2005). Based on RDA results, pesticide toxicity in terms of $sumTU_{P.\ subcapitata}$ was the major driver of these differences. The toxicity can be mainly attributed to fungicides because insecticides exhibited a minor contribution to the sumTU. The results of the laboratory experiment support that fungicide toxicity causally contributed to the fungal community change observed under field conditions. For example, *Clavariopsis aquatica* decreased with increasing toxicity during both the field and the laboratory experiment. Besides, our field results (Fig. 2a) are in general agreement with a study (Bundschuh et al., 2011b) reporting that the leaf-borne fungal species *Alatospora acuminata*, *C. aquatica* and *Flagellospora curvula* were reduced by the fungicide tebuconazole, whereas *Tetracladium marchalianum* exhibited no response to this toxicant. Nevertheless, some patterns found in the field deviated from the response in our laboratory experiment. For example, *Heliscus lugdunensis* showed opposite responses in the field study (Fig. 2a) and laboratory experiment (Fig. 2c). The positive relationship between conidia abundance of *H. lugdunensis* and fungicide toxicity found in the laboratory was also reported in an experiment with the fungicide cyprodinil (Zubrod et al., 2015), which was also included in our experiment. Generally, differences between the field situation and laboratory experiments are not limited to fungicides and have been reported for several stressors (Schäfer, 2014). Explanations include different exposure patterns and recovery occurring in the field compared to our short-term experiment and field communities are shaped by the environmental context, i.e. a wide range of abiotic and biotic variables (Liess and Beketov, 2011).

Beside changes in fungal community composition, fungicide toxicity was associated with a decrease in fungal biomass and an increase in the density of bacteria. In linear modelling that included additional explanatory variables, fungicide toxicity was the only predictor of fungal biomass, whereas multiple physico-chemical and habitat variables, dominated by phosphate, explained the increase in bacteria. These results suggest that indeed fungicide toxicity affected fungal communities including their biomass, whereas the increase in bacteria may be related to elevated nutrient levels. Moreover, the decrease in fungal biomass along the fungicide toxicity gradient may have contributed to the decrease in $k_{microbial}$ (Gessner and Chauvet, 1994), which reached up to 40% reduction compared to sites with no or minor fungicide exposure. However, this explanation relies on the assumption that fungi are more important than bacteria for LD, which has been shown in several studies (Duarte et al., 2010; Pascoal and Cássio, 2004), though similar importance of fungi and bacteria has been reported

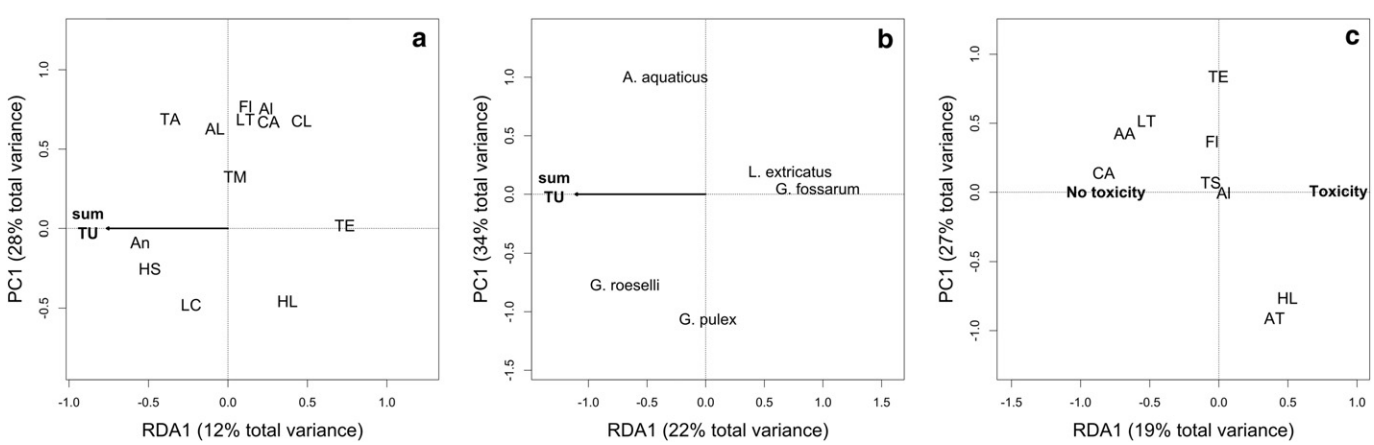

Fig. 2. a) Triplot from the redundancy analysis using fungal species (field experiment) as response variables and the variables selected by the automatic model building as explanatory variables. Species: *Alatospora* sp. (Al), *Anguilospora longissima* (AL), *Anguillospora* sp. (An), *C. aquatica* (CA), *C. longibrachiata* (CL), *Flagelospora* sp. (Fl), *H. lugdunensis* (HL), *Heliscella stellata* (HS), *L. curvula* (LC), *Lemonniera terrestris* (LT), *Tricladium angulatum* (TA), *T. marchalianum* (TM) and *T. elegans* (TE). b) Triplot using invertebrate species (field experiment) as response variables and the variables selected by the automatic model building as explanatory variables. c) Triplot from the redundancy analysis using fungal species (laboratory experiment) as response variables and fungicide toxicity after partialling out pH effect as explanatory variable. Species not present in Fig. 2a: *Articulospora atra* (AA), *Articulospora tetracladia* (AT) and *Unknown* sp. (Un).

occasionally (Hieber and Gessner, 2002). It remains speculative whether the increase in bacterial density of up to a half logarithmic unit in sites with high fungicide toxicity partially buffered the decrease in $k_{microbial}$.

Despite the fact that fungicide toxicity exhibited the highest correlation with several endpoints related to the microbial decomposer community, pH and nitrite exhibited the strongest explanatory power for $k_{microbial}$ (Table 2), displaying a negative relationship. The potential effect of pH on freshwater fungi, and subsequently on $k_{microbial}$, has been suggested to be indirect (by affecting for example the solubility of metals; Bärlocher, 2005). Nitrite was significantly correlated with conductivity and the percentage of dissolved oxygen ($r = 0.61$ and -0.60, respectively). Both increasing conductivity and decreasing dissolved oxygen have been reported to negatively affect $k_{microbial}$ (Pascoal and Cássio, 2004; Schäfer et al., 2012a). Overall, the results suggest that fungicides were the main driver of the effects on microbial-community related endpoints, whereas shifts in the microbial decomposition rate were largely driven by physico-chemical variables. This mismatch, i.e. that the changes in the ecosystem structure and processes were related to different stressors, can be explained by (1) the integrative nature of ecosystem processes that are influenced by a magnitude of environmental variables (Tank et al., 2010) and by (2) the fact that the response of ecosystem processes depends rather on changes in the functional composition of communities than on the taxonomic composition (Vandewalle et al., 2010). Moreover, complex fungicide mixtures occur in the field and their effect on ecosystem processes may not be captured with the very simplified toxic unit approach. For example, the sumTU assumes additive mixture effects (concentration addition), whereas non-additive effects of fungicide mixtures can occur (Zubrod et al., 2015). In addition, the sample size in our study was relatively low and the results from models with multiple variables should be interpreted with caution and against the results from previous studies (Rasmussen et al., 2012b; Schäfer et al., 2012a). Future studies should try to unravel the effect mechanisms using a combination of experiments under controlled conditions and field surveys covering a larger set of sites.

4.2. Effect of fungicides on shredder community and invertebrate LD

Fungicide toxicity in stream water was not related to $k_{invertebrates}$, lending no support to the hypothesis that fungicide toxicity lead to a reduction in this ecosystem process. This contrasts with our expectations of finding an indirect effect on $k_{invertebrates}$, as fungal species preferred by gammarids were more abundant in non or low-polluted sites, which in turn indicates that food intake may not be affected by the fungal community composition if no alternative food is present (e.g. *Alatospora* sp. or *C. aquatica*; Arsuffi and Suberkropp, 1989; Jabiol and Chauvet, 2012). The absence of detectable effects may be explained by the fact that measured fungicide concentrations were below effect thresholds for gammarids (Zubrod et al., 2014, 2015), together with the temporal mismatch between pesticide sampling and LD assessment. Nonetheless, gammarid feeding measured in situ was similarly unrelated to fungicide toxicity either, despite being measured in concert with pesticide sampling. Besides, no relationship between gammarid feeding and $k_{invertebrates}$ was found (Supplementary data Fig. S6), which may also be attributed to the temporal mismatch.

Our results contrast with previous field studies reporting a decrease in $k_{invertebrates}$ because of the loss of pesticide-sensitive species (Schäfer et al., 2012a, 2012b). However, these studies reported data from agricultural areas with a more diverse shredder community, whereas a high dominance of gammarids was found in our study area. In addition, the present study is in agreement with Rasmussen et al. (Rasmussen et al., 2012b) where a *G. pulex*-dominated shredder community in Danish agricultural streams was unrelated to a pesticide gradient of a comparable TU range. Rasmussen et al. (2012b) found a positive correlation of $k_{invertebrates}$ with the density of gammarids, whereas no significant correlation was found in our study between $k_{invertebrates}$ and total gammarid abundance, though $k_{invertebrates}$ tended to be higher at higher gammarid abundances (Supplementary data Fig. S7). Nonetheless, the fact that in our study LD and decomposers were sampled in autumn implies that (i) insect shredders may have emerged and that the shredder community can be more heterogeneous and potentially more sensitive during other seasons, and (ii) the exposure to fungicides was lower than in summer (Fernández et al., 2014). Therefore, effects on $k_{invertebrates}$ from recurrent fungicide exposure could occur earlier in the year as has been shown in Schäfer et al. (2012b). Indeed, the distribution of the three dominant gammarid species was associated with waterborne fungicide toxicity, although the latter correlated with several other environment variables. Moreover, both $k_{invertebrates}$ and the gammarid feeding rate were negatively associated with sediment-borne copper concentrations. Decreasing abundance and diversity of crustaceans and other macroinvertebrates have been reported in other studies at sediment copper concentrations detected also during the present study (Kraft and Sypniewski, 1981; Mebane, 2002). Nonetheless, the association should be interpreted with caution because sediment copper toxicity for aquatic organisms is governed by the availability of free copper ions in the water column (Kramer et al., 2004), which in turn depends and dissolved organic carbon and pH (De Schamphelaere and Janssen, 2004).

5. Conclusions

Our results suggest that recurrent exposure to fungicides in stream water affects microbial communities. However, whether these changes result in a decrease in $k_{microbial}$ remains speculative and requires further investigation. Both field studies and experiments under controlled conditions are required to disentangle the effects of environmental variables and fungicide toxicity on microbial decomposers and LD. $k_{invertebrates}$ was negatively associated to sediment copper, whereas no response to fungicide toxicity in water was observed.

Author contributions

Idea and study design: RBS, MB and DF; performance of field study and laboratory experiments: DF, KV and JPZ; data analysis: DF and RBS; manuscript drafting: DF and RBS; manuscript revision: all.

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Johanna Höger assessed gammarid feeding rate. Natalia Koksharova assisted during conidia identification. Sebastian Schadt identified invertebrate samples. Natalie Kolb und Anne Heuring assisted in the field. Zacharias Steinmetz and Kilian Kenngott processed copper samples. Jochen P. Zubrod was funded through a scholarship of the German Federal Environmental Foundation (Deutsche Bundesstiftung Umwelt) during the conduct of this study. The Rhineland Palatinate fishing association provided facilities used at different stages of this study. The study was funded by the German Research Foundation (DFG; grant number: SCHA 1720/3-1).

Appendix A. Supplementary data

The supplementary data associated with this article contains a R script for reproducing the data analysis and a document with the following information regarding the field study: sampling schedule (Fig. S1), location of the study area and distribution of sampling sites (Fig. S2), fungicide concentrations in stream water (Table S1), relationship between toxicity metrics and biotic endpoints (Table S2), relationship between fingicide toxicity and the endpoints invertebrate LD and gammarid feeding rate (Fig. S3), environmental variables with highest explanatory power for LD when retaining copper as predictor variable (Fig. S4), contribution of gammarids to the total shredder abundance

(Fig. S5), relationship between invertebrate LD and gammarid feeding rate (Fig. S6) and relationship between gammarid abundance and invertebrate LD (Fig. S7). This material is available free of charge via the Internet at STOTEN TO http://dx.doi.org/10.1016/j.scitotenv.2015.06.090.

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